

ACTIVITY 2 — LEARNER WORKSHEET**Measure — Map the Process and Spot the Waste**

TYPE	DURATION	MAPS TO	TOPIC
Hands-on Lab	30 minutes	MEASURE	Measure — See What Is Really Happening

Name	
Date	

What you are doing

The team now has an agreed problem statement, but still only opinions about where the time goes. Management wants to act immediately; the team has persuaded them to spend one morning measuring first.

What you will produce: A completed SIPOC, a 15-step process map with each step judged VA or NVA, a completed DOWNTIME waste tally, and the written baseline figures.

The scenario

The improvement team spent one weekday morning observing the 7:30am-9:00am rush with a stopwatch and a check sheet. They timed 60 customers end to end, recorded what each one ordered, how long they waited, whether the drink had to be remade, and what was visibly holding things up. They also timed every step of the process itself.

Your job is to turn that raw observation into a picture of the process as it really runs — not as the staff handbook describes it — and to find where the time and effort are actually being lost.

Your steps

1. **Agree the start and stop points** — Fix the boundaries before anything else: start when the customer joins the queue, stop when the customer receives the drink. A process with no agreed boundaries produces a map nobody can compare against anything.
2. **Complete the SIPOC** — Fill templates/sipoc-template.csv: Suppliers, Inputs, Process, Outputs, Customers. Hold the Process column to 5-7 high-level steps — the detail belongs in the process map, not here. A SIPOC is the one-page view a manager can read in thirty seconds.
3. **Build the process map with real timings** — Open data/process-step-timings.csv — the 15 observed steps with their average times. Copy them into templates/process-map-template.csv. Note the 'Times Step Was Skipped' column: step 10 (fetching milk) happened for 18 of 60 customers and step 15 (remaking a drink) for 7 of 60.
4. **Judge every step value-adding or not** — Mark each step VA or NVA and write one line saying why. The test is applied per STEP: would the customer pay extra for THIS step? Not

'does the customer want coffee'. Expect to find that the great majority of the elapsed time is non-value-adding — that is the normal and useful finding.

5. **Classify the data you have** — Look at the columns in data/morning-rush-observation-log.csv. Which figures are discrete (counted — remakes, tallies) and which are continuous (measured on a scale — wait time in minutes)? Note that 'Drink Ordered' and 'Delay Reason' are neither: they are categorical labels.
6. **Complete the DOWNTIME waste tally** — Open data/downtime-waste-tally.csv — three rows are completed as worked examples. Using the observation log and the step timings as your EVIDENCE, complete the remaining five rows in templates/downtime-tally-template.csv. Some waste types may legitimately be zero here; recording zero with evidence is a correct answer.
7. **Write down the baseline** — Record the average wait, the longest wait, the percentage of customers over the 5-minute promise, and the number of drinks remade. Identify which single waste type you tallied most often and at which step. Activity 5 measures the improvement against these numbers, and the Case Study expects a baseline to be quoted rather than invented.

Your working space

Complete every field. You may keep this worksheet and refer to it during the open-book assessment.

Step 1 — Agree the start and stop points

Step 2 — Complete the SIPOC

Step 3 — Build the process map with real timings

Step 4 — Judge every step value-adding or not

Step 5 — Classify the data you have

Step 6 — Complete the DOWNTIME waste tally

Step 7 — Write down the baseline

Answers to the discussion prompts

1. Looking at your process map, what proportion of the total time is genuinely value-adding — and does that surprise you?

2. Which single step would you attack first, and is that step a symptom or a cause?

3. Which waste types came out at zero, and what evidence lets you say so with confidence?

4. Why does the team need to agree what is recorded, who records it and when, BEFORE collection starts?

Reflect & discuss

Was the step you assumed was the bottleneck actually the one the data pointed to?

How would the map differ if you had asked staff to describe the process instead of observing it?

Which of these wastes do you recognise in your own workplace?

Check your work

Every group can show a SIPOC with all five columns filled, a process map with every step timed and judged VA or NVA, a complete DOWNTIME tally tied to evidence, and the written baseline figures.